Shanmuga Ganesh

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SUMMARY

Senior Full-Stack AI and Data Engineer with **8+ years** of experience building scalable **enterprise, fintech, healthcare, and AI platforms** with **Java**, **Python**, **PySpark**, **React**, **Next.js**, and **AWS**. Leads engineering teams on Sherwin-Williams' **AI-SDLC platform** for **2K+ learners**, backed by **Databricks Medallion pipelines**, **Spark performance tuning**, and an **entity matching service** that resolves multi-source records into **canonical IDs**. Built a **security master (MDM)** for fintech market data and **clinical research data pipelines** with **record linkage**, **deduplication**, and **data quality scoring**. Strong in **distributed systems**, **big data engineering**, **customer-facing APIs**, and **technical leadership**.

SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, SQL
- **Frontend:** React, Next.js, Angular, HTML5, CSS3
- **Backend and APIs:** Spring Boot, Spring Security, Hibernate, Django REST, FastAPI, Node.js, Celery, REST, GraphQL, WebSockets, API Versioning, Prisma, Drizzle ORM
- **Data Engineering:** Databricks (Lakeflow Jobs, SQL, Dashboards, Unity Catalog), Apache Spark, PySpark, Spark SQL, Spark **Performance Tuning**, Delta Lake, Medallion Lakehouse, ETL/ELT, Batch and Incremental Ingestion, Configurable Data Intake, Parquet, Avro, Data Modeling, Data Quality, Data Quality Scoring
- **Entity Resolution and MDM:** Entity Matching, Record Linkage, Deduplication, Fuzzy Matching, Canonical ID Assignment, Master Data Management (MDM), Reference Data Management
- **Databases and Messaging:** PostgreSQL, Lakebase, MySQL, MongoDB, DynamoDB, Redis, Kafka, RabbitMQ, Indexing, Transactions, Query Optimization
- **AI and ML:** PyTorch, CUDA, scikit-learn, GPT, BERT, RAG, LLM Workflows, Prompt Engineering, OpenAI APIs, ML Pipelines, Anomaly Detection
- **Cloud and DevOps:** AWS, Azure, Docker, Kubernetes, GitHub Actions, CI/CD
- **Security and Testing:** OAuth 2.0, OIDC, PKCE, JWT, RBAC, IAM, Microsoft Entra ID, Better Auth, SSO, PyTest, JUnit, React Testing Library, Integration, E2E, API and Performance Testing
- **Tools:** Git, GitHub Copilot, Claude, OpenAI Codex, SonarQube, SonarCloud, ServiceNow, Jira
- **Architecture:** Microservices, Distributed Systems, Event-Driven Architecture, System Design, Technical Leadership, Design Reviews, Mentoring

EXPERIENCE

Senior Software Engineer - FullStack AI FDE

Sherwin-Williams **August 2026 - Present, Cleveland, OH**

- Led a team of **5 engineers** on Cover the Codebase, an **AI-SDLC platform** for **2K+ learners**, owning **design reviews**, **code reviews**, and **pipeline standards** while building **configurable data intake** across GitHub, Copilot, Claude, Codex, SonarQube/SonarCloud, ServiceNow, and Jira.
- Built an **entity matching service** resolving **2M+ records** from GitHub, **Entra ID**, Pluralsight, and Copilot into **canonical learner IDs** using deterministic and **fuzzy matching**, **deduplication**, and **identity mapping**, achieving a **98% match rate**.
- Architected **Databricks Medallion pipelines** processing **1.3M+ daily events** with **PySpark**, **Delta Lake**, **Unity Catalog**, and **Lakeflow Jobs**, applying **Spark tuning** (partitioning, **broadcast joins**, **skew handling**, **Z-ordering**) to cut runtime by **33%** across **23 adoption metrics**.
- Designed **Lakebase PostgreSQL** schemas with **data quality scoring**, **status assignment**, and **correction-aware upserts** that **sync resolved entities** to downstream dashboards, load-tested at **5.4M rows** and **2K concurrent queries**.

Software Engineer (Healthcare Research)

Case Western Reserve University **March 2026 - July 2026, Cleveland, OH**

- Built an **AI-assisted quality system** for **clinical microbiome research data**, applying **feature validation**, **statistical filtering**, and **metadata anomaly detection** to reduce false-positive carryover by 35% across **32K+ samples (48+ TB)** from the **Human Microbiome Project (HMP)**.
- Engineered a **Python HPC sequencing pipeline** with **QIIME2** and **DADA2** for **clinical sample ingestion**, **quality control**, and **taxonomy profiling**, cutting analysis turnaround by **30%** through scalable, reproducible execution.
- Standardized **clinical metadata** and **sample records** across research cohorts with **validation rules**, **deduplication**, and **automated quality checks**, improving **pipeline reproducibility by 40%** and producing **ML-ready healthcare datasets**.

Software Engineer (Fintech)

Musk and Gale **June 2025 - Present, Cleveland, OH**

- Led a **3-engineer team** on a **Java** and **Python Market Data Aggregation Platform**, owning **architecture** and **design reviews**, and built a **security master (MDM)** matching **3.1M+ instruments** across **40 vendor feeds** by **CUSIP**, **ISIN**, and ticker into **canonical IDs**.
- Built **Django REST APIs** and **React** interfaces powering **customer-facing** and **internal self-service tools**, reducing **API latency by 32%** through **query optimization**, payload tuning, and cleaner service calls.
- Engineered **configurable ingestion** with **WebSockets** and **Celery**, onboarding new vendor feeds through **schema reconciliation** without code changes, and processing **1.8M records per hour (43M+ daily)** with **retry handling** and **fault tolerance**.
- Designed **PostgreSQL** data models over **45M+ rows** of financial data with **validation** and **quality scoring** rules, improving **query performance by 41%**, and and **set data-modeling standards** for the team.

MTS (Full Stack Software Developer - CRM)

ZOHO Corporation **January 2022 - December 2023, Chennai, India**

- Revamped the **Tax Configuration** system using **Java** and **API versioning**, enabling backward-compatible tax workflows across **7+ integrated systems** with scalable design and reliable release management.
- Built a **Java NIO Custom Validation Rules** system for CRM record creation and edit flows, processing **1.47+M records per day** with **non-blocking validation** and high-throughput execution.
- Built a **Kafka-based Rollup Summary pipeline** to update parent aggregation fields from child record changes, processing **13+ M record events per day** with scalable **event-driven architecture**.
- Rebuilt Zoho CRM **Recent Items storage** to cache each user's latest 20 records, reducing **retrieval latency by 50%** and improving low-latency record navigation.

Junior Associate (Full Stack Software Developer)

Augusta Hitech Software Solution **November 2018 - December 2021, Coimbatore, India**

- Developed **Angular** and **Node.js CRM modules** for marketing and sales workflows, improving customer engagement by **32%** through reusable UI components, scalable APIs, and workflow automation.
- Built **Node.js lead allocation workflows** with **Angular dashboards**, increasing qualified lead retention by **60%** through automated routing, real-time visibility, and improved sales execution.

EDUCATION

Master's in Computer Science

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